

Tarun Vuppala

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SUMMARY

Software Engineer at the start of my career, with experience on professional engineering teams and in personal projects. I break problems down, think through trade-offs, and have developed an intuition for the problem at hand and what it requires, independent of prior knowledge.

INTERNSHIP

Unity Internet

March 2025 - June 2025

Software Developer Intern

Hyderabad, India

- Built a Premiere Pro auto-edit plugin that reduced podcast editing time from 3 hours to under 8 min, using audio signal analysis, real-time scene switching, and overlap handling.
- Shipped cross-platform binaries adopted by 10+ podcast editors, reducing manual release overhead by 60%.

Front Interactive Services

June 2024 - February 2025

Software Developer Intern

Hyderabad, India

- Engineered backend systems with 50+ REST API endpoints and a scalable server-side architecture.
- Worked in an Agile Scrum team of 6 engineers across 2-week sprints using TDD, reducing post-release defects by 30% and improving system stability.

PROJECTS

Atlas AI - React Native, llama.cpp, Python, RAG | Live Demo

Feb 2026 - Apr 2026

Offline Multi-Expert AI System

- Built an offline-first AI mobile app with on-device LLM inference using llama.cpp and a custom RAG expert system; chunked and embedded domain knowledge into a vector database to enable private, and context-aware conversations.
- Controlled memory footprint using GGUF quantization and persistent context windows, with per-device parameter tuning that kept stable mobile TTFT between 1.5-8s.

Acethletics - Next.js, Node.js, MongoDB, WebSockets | Live Demo

Aug 2024 - Dec 2024

Centralized Athletic Sports Platform

- Architected backend systems and database schemas that scaled to 2,000+ downloads and 1,500 peak concurrent users during live tournament usage, maintaining 100ms average API response latency across 10+ live events.
- Engineered WebSocket infrastructure delivering 200+ real-time events per match at 20ms propagation latency, with zero downtime recorded across all tournaments.
- Implemented transactional workflows with zero production data inconsistencies and standardized error handling and HTTP response across all endpoints.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python (FastAPI, Flask), JavaScript/TypeScript (React, Next.js, Node.js), Java (Spring Boot), Go, C++, C, SQL, GraphQL, REST, WebSockets
- **Backend & Systems:** System Design, Microservices, API Gateway, gRPC, Message Queues (Kafka, RabbitMQ), Caching (Redis), JWT, OAuth2, Cron Jobs
- **AI/ML:** Deep Learning (Transformers, CNNs), LLMs (OpenAI, DeepSeek, Llama), RAG, Vector DBs (ChromaDB, LanceDB, pgvector), Hybrid Search, Re-ranking, Feature Engineering, PyTorch, TensorFlow, Hugging Face, Ollama
- **Databases & Data Engineering:** PostgreSQL, MySQL, MongoDB, Pandas, NumPy, Schema Design, Query Optimization, Index Tuning
- **Cloud & Infra:** AWS (EC2, Lambda, S3, RDS, Bedrock), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Monitoring (Prometheus, Grafana), Linux, Shell Scripting, Git, pytest, Postman

EDUCATION

ACE Engineering College

Hyderabad, India

Bachelors of Technology (B. Tech) in Computer Science

Nov 2022 - Apr 2026

EXTRA- CURRICULAR

- **TEDx:** Co-organizer, TEDxACEEC '25; handled speaker outreach, secured sponsorships, and event coordination.
- **AICTE Yukti 2023:** National Finalist in YIC presenting an innovative startup concept.
- **Published Author (IJVRA, 2026):** Developed the underlying backend and LLM integration framework for "ATLAS AI: Offline Multi-Expert AI System with Edge Device Computation." [Paper]
- **GATE 2025:** Qualified GATE 2025 in Computer Science.